

AMEE 2026 · VIENNA

An Integrated Core-and-Cluster Framework for Teaching Physical Examination

Aligning examination technique with clinical reasoning

R. P. Soriano · J. Hojsak · M. Herscher · B. Storch · B. Forsyth · C. Strother

Climb higher. Reach further. Care deeper.

Leni and Peter W. May Department of Medical Education · Icahn School of Medicine at Mount Sinai



Disclosure Statement

Conflicts of interest

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Three approaches to teaching

Head-to-toe

The learner decides

The sequence is set in advance, so selection is not part of the task.

Reported

Criticized for teaching maneuvers without the context that makes them meaningful, and for delivering all of it at once.

Wilkerson & Lee 2003 · Haring 2014

Core + Cluster

The learner decides

Which cluster the presentation indicates.

Reported

Questioned on how many clusters a curriculum needs, and whether a focused examination risks missing findings when symptoms are vague.

Gowda 2014 · Jewett 2024

Hypothesis-driven

The learner decides

Which single maneuver tells two diagnoses apart.

Reported

Validity evidence for the method, but it asks for diagnostic reasoning beyond early learners.

Yudkowsky 2009

Increasing demand on clinical reasoning



What Core and Cluster is

One examination, assembled from two parts, for every patient encounter

Part 1 · The core

Performed for every patient

A fixed set of maneuvers done whatever the patient presents with. It does not change from encounter to encounter.

Part 2 · One cluster

Selected by the presentation

An additional set of maneuvers for that clinical domain. The presentation determines which one, and only one is used.

Every encounter is the core plus one cluster. Clusters are alternatives, never added together, so the examination is never the whole textbook.

Chest tightness

core + cardiac cluster

Dizziness

core + neuro-vestibular cluster

Finding 1 — before the framework

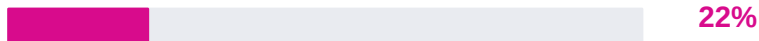
Selection did not follow the case

Simulated patient encounters, before the framework

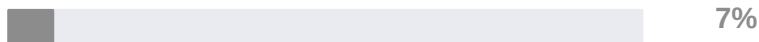
Left out what was indicated

Headache case — not performed at all

Indicated neurologic maneuvers



Maneuvers done for every patient



The maneuvers a headache calls for were the ones most often skipped.

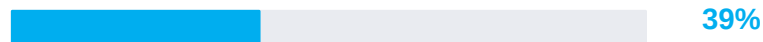
Put in what was not

Heart and lung case — jugular venous pressure

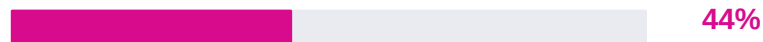
Attempted the maneuver



Performed it to standard



Mispositioned the patient



Nearly always done. Correct in fewer than four cases in ten.

In both cases, the maneuvers performed were not the ones the case called for.

How we adopted and specified the model

5 faculty who lead clinical skills teaching, the preclerkship and clerkship curriculum, standardized patients, and simulation

Adopted

Core + Cluster

Gowda 2014 national survey

Jewett 2024 early-learner adaptation

Gowda 2026 toolkit — 10 years across several schools

Specified ourselves

- The maneuvers forming our core
- The 8 clusters and their presenting problems
- One cluster per presentation, not several

Gowda 2026: schools may modify the core to fit their curriculum.

What we did not change

The performance descriptors. Our school's written standards for full and partial credit were adopted unchanged, so the framework changed which maneuvers a student performs — not the standard they are held to.

That matters for the comparison later in this talk.

The Core examination

24

examination
maneuvers

in seven sections

1 General appearance and vital signs

2 Head, neck and cranial nerve screen

3 Cardiac

4 Pulmonary

5 Abdomen

6 Vascular

7 Neurologic screen

3 universal items
hygiene before and after · draping

The three universal items are performed on every patient whatever the complaint. They are not examination maneuvers, so they sit outside the core count — which makes 24 directly comparable with other published cores.

Based on the national core described by Gowda and colleagues (2014); Jewett and colleagues (2024) adapted the same core for preclinical students.

Eight clusters and their presenting problems

Cardiac

Chest pain · palpitations · syncope · edema

Gastrointestinal

Abdominal pain · nausea · diarrhea · bleeding

Musculoskeletal

Joint, back and neck pain · limb swelling

Genitourinary

Dysuria · flank pain · hematuria · pelvic pain

Pulmonary

Dyspnea · cough · hemoptysis · wheeze

Neuro-vestibular

Dizziness · vertigo · weakness · imbalance

Head, ears, eyes, nose, throat

Sore throat · ear pain · red eye · congestion

Endocrine and systemic

Fatigue · weight change · temperature intolerance

The presenting problem determines which cluster is used. Clusters are alternatives, not a cumulative list.



How we taught it

Choosing a cluster is a reasoning task, so we taught the reasoning alongside the maneuvers

Semester 1

The full head-to-toe examination, to fluency

Semester 2

Core and Cluster. Each cluster timed to the organ-system course running in parallel

Before the clerkships

Tested on the maneuvers the case indicates

this is what we measured

Two kinds of practice

Standardized patients

performing the cluster on a real encounter

Paper cases

differential diagnosis skills sessions

What we asked in those sessions

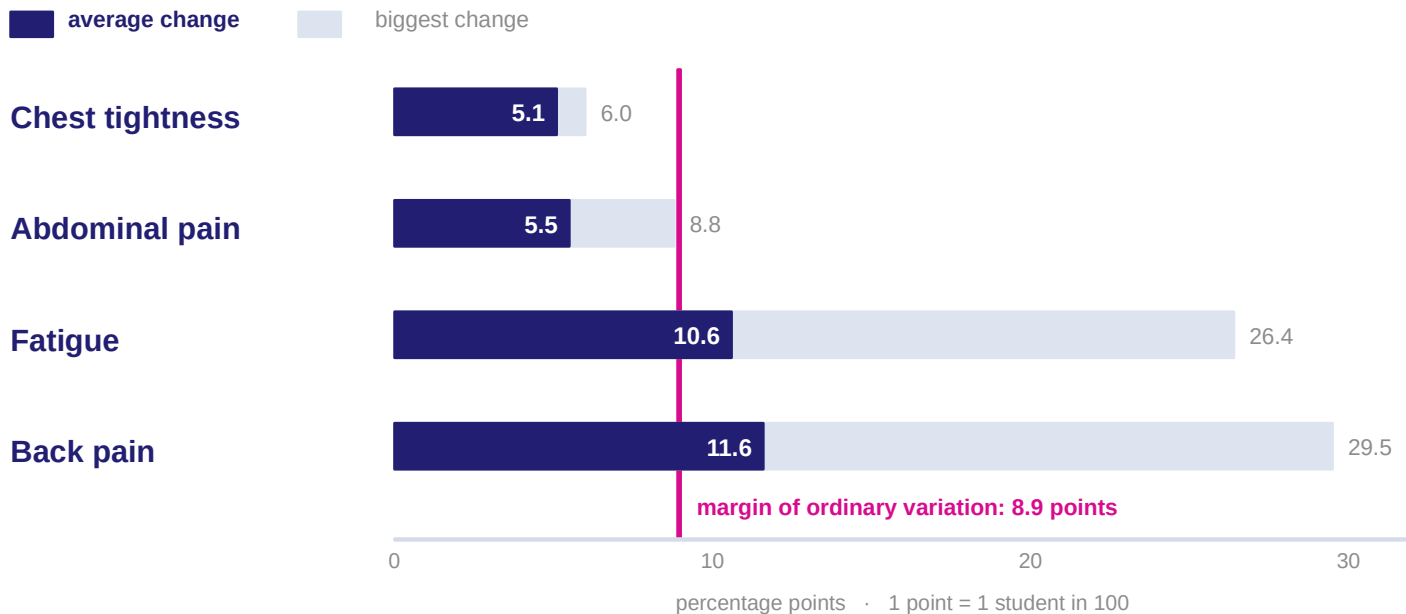
- What do you think is going on?
- What would you examine, and what are you looking for?
- How would the findings change your thinking?

The reasoning sequence a cluster is built on is described by Gowda and colleagues (2026).

Finding 2

Scores change from year to year

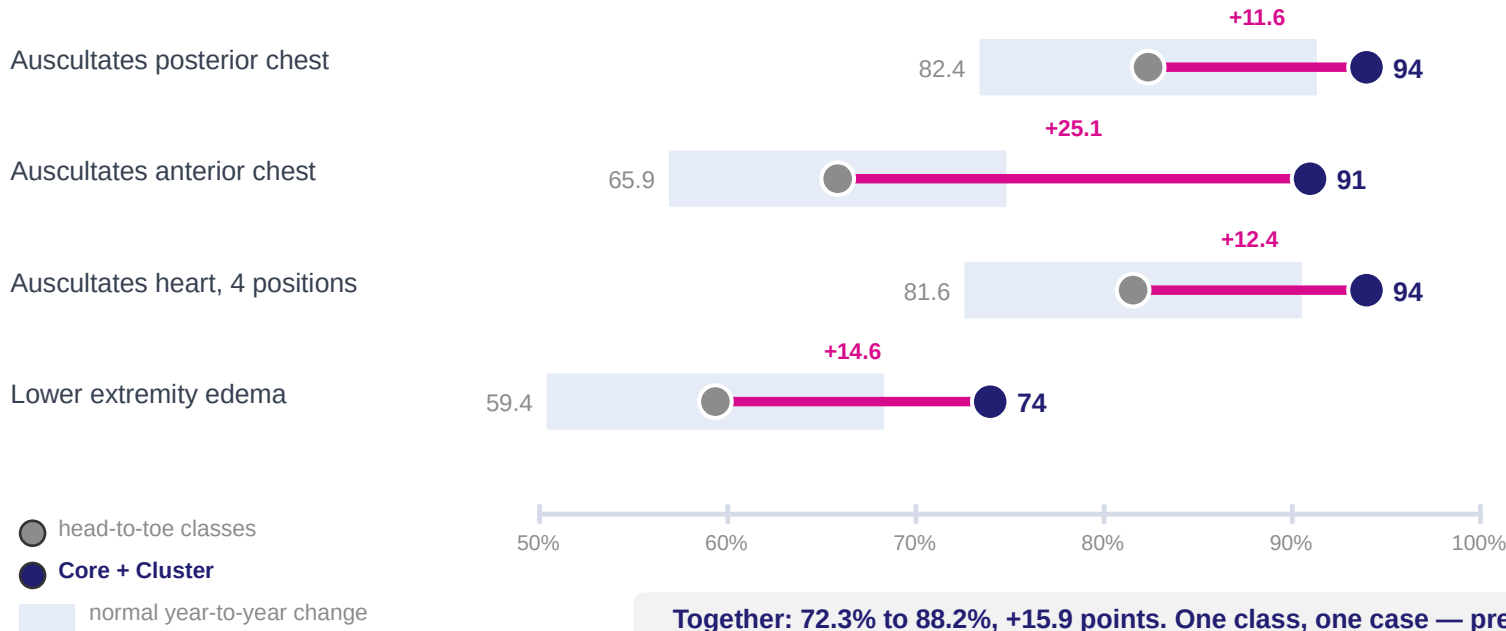
Same teaching, 3 classes in a row. How far apart the best and worst year were.



Finding 3

The maneuvers this case called for

4 examination maneuvers this case indicates, scored the same way in both years. Percentage who did each one completely.



What this design cannot show

Four things this comparison cannot tell us

We added it, we did not start with it

All 4 classes learned the full head-to-toe exam in semester 1. We changed only the second semester. The classes were also in different years.

We measured what they did, not why

A student could pick the right cluster from memory. And we scored completeness, not accuracy — more maneuvers can mean more false positives.

One case, one body system

One heart and lung case, and 7 maneuvers. We scored nothing outside that system, so we cannot say what students left out.

Agreed by our faculty, not tested elsewhere

Our faculty decided which maneuvers go in the core and which cluster each problem needs. No one outside has checked that.



Conclusion

What we conclude

- 1 Core and Cluster is a workable framework for teaching and assessing the physical examination.**
24 core maneuvers, 8 clusters keyed to presenting problems, and written performance criteria for every maneuver.
- 2 It appears to improve how completely students perform the examination a presentation indicates.**
Preliminary, from one class: compared with 3 head-to-toe classes, by more than these scores vary between classes on their own.
- 3 The framework is built to be adapted.**
Any program can take the core, the clusters, or the performance criteria, in whole or in part.

Next

Do students omit what is not indicated?

an instrument scoring beyond one system

Does it work as a foundation?

the core-first class sits the same exam next year

Can the core be smaller?

a stated criterion, rated by faculty



References

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Where the evidence stands

All three positions have been argued.

The concerns on each approach have been in the literature since the model appeared.

The only published comparison found no difference — on a measure taken two years after instruction, which its authors attribute to washout of the teaching effect.

We are not going to settle it today. What we can offer is one school's implementation, and some early data.

Jewett L, Clarke S, Griffin E, Danielson A. BMC Med Educ. 2024;24:337.



Two things we set out to do

1 Build it

1

Adopt Core and Cluster — a fixed core performed for every patient, plus one cluster selected by the presenting problem — and specify it well enough to teach and to assess.

2 Find out whether it changed anything

2

Compare students taught this way with earlier classes taught head-to-toe.

Where the model sits, following Kelly and colleagues (2020)

Head-to-toe

First year

Core + Cluster

Early second year

Hypothesis-driven

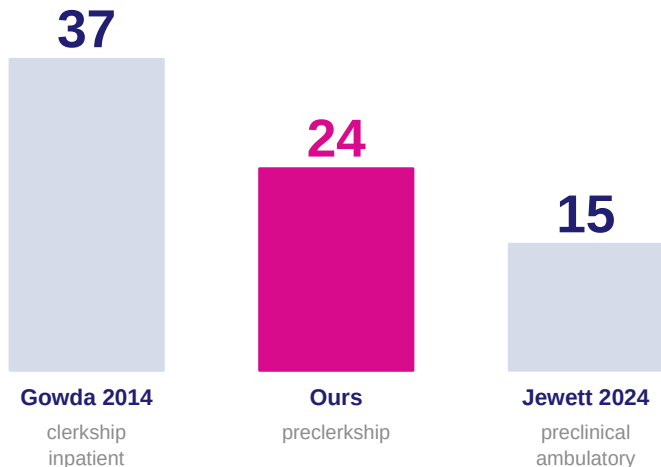
Later

In the class measured, Core + Cluster replaced a second-semester review of the head-to-toe examination.



How we arrived at twenty-four

Examination maneuvers only. Hygiene and draping excluded from all three counts.



7 maneuvers moved, not deleted

Gowda core items that sit in our clusters

Cardiac

Jugular venous pressure · carotid palpation · auscultation at apex

Gastrointestinal

Liver palpation

Endocrine

Thyroid palpation

Neuro-vestibular

Deep tendon reflexes · sensory testing

Gowda's core was built for clerkship students admitting inpatients. Ours is for first-year students.

Structure of each cluster

Every cluster is built the same way

A

Core maneuvers relevant here

restated, so learners see which carry diagnostic weight

B

Maneuvers added for this domain

each with FULL and PARTIAL descriptors

3 to 20

+

Optional add-ons

equipment, chaperone, or wider scope of practice

programs may omit

Example

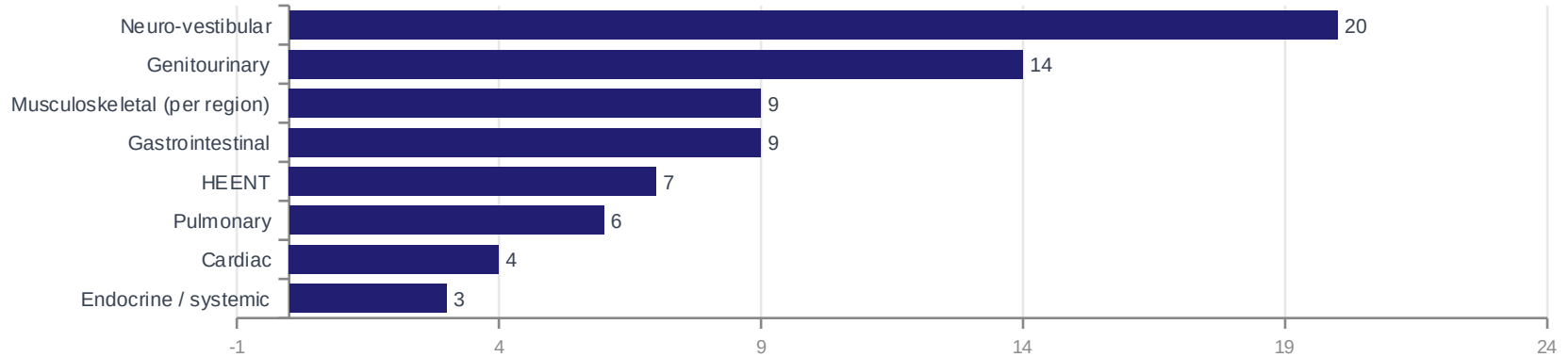
Neuro-vestibular

A vital signs · cranial nerves · motor strength · gait

B reflexes · Romberg · cerebellar testing · vibration · proprioception



The framework in one line: 24 + 1



Maneuvers each cluster adds to the 24-manoeuvre core. Musculoskeletal is organized by region. Optional add-ons excluded.



Ours and Jewett's, side by side

Both adapted Gowda for preclerkship learners, for different settings.

In ours, not in Jewett's

9 maneuvers

Vital signs

blood pressure, heart rate, respiratory rate

Orientation

person, place, time

Chest percussion

posterior, 3 levels

Abdominal auscultation

before palpation

PMI palpation

Jewett auscultates but does not palpate

Light and deep palpation

scored as two items, not one

In Jewett's, not in ours

nothing dropped outright

"Sick / not sick" appearance

absorbed into our general appearance

Hip flexion against resistance

absorbed into LE motor strength

Forearm flexion and extension

absorbed into UE motor strength

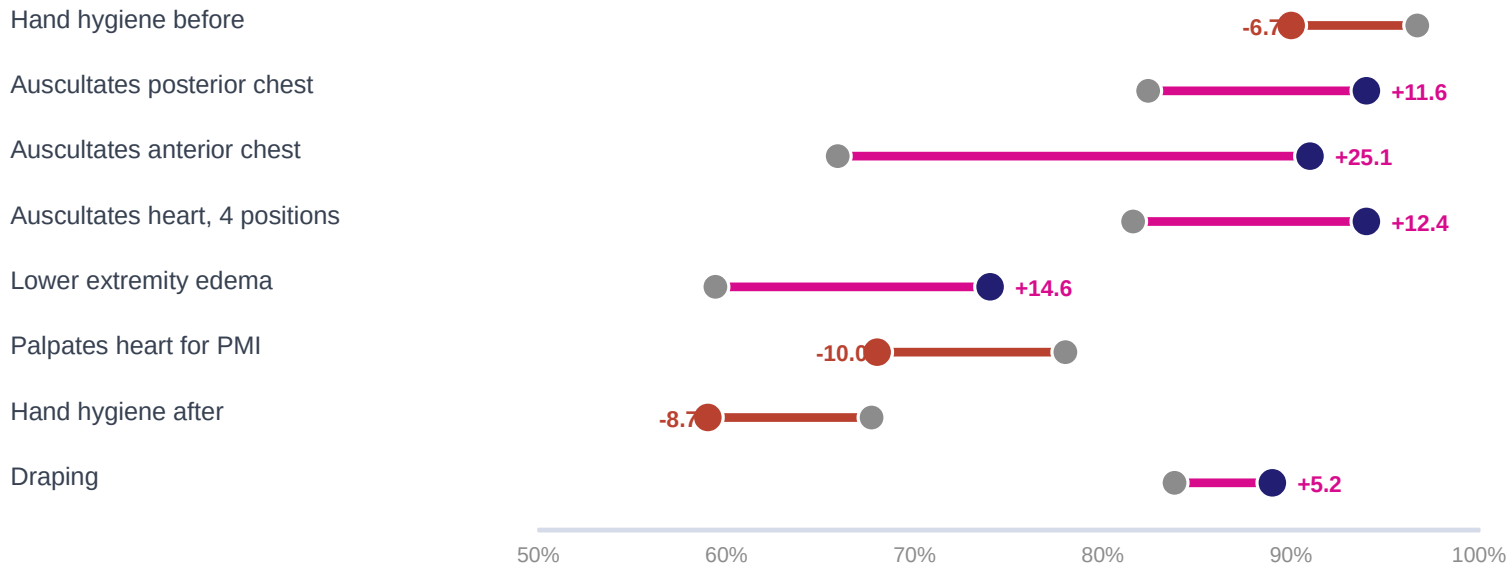
Jewett's core is a subset of ours. The difference is a well-patient screening exam versus one built to carry a presenting problem.

Gowda 37 (inpatient admission) · ours 24 (complaint-based preclerkship) · Jewett about 15 (well-patient ambulatory). Different settings, not points on one scale.



All 8 matched maneuvers

Including the 3 universal items, which the framework counts separately.



All 8 together: 76.9% to 82.4%, a gain of 5.4 points — inside the normal range.

Selection, not execution

Core + Cluster cohort, cardiopulmonary station, 110 students, 11 maneuvers

Execution, among those who attempted

91 to 95 percent complete on ten of the eleven maneuvers.
Jugular venous pressure is the exception at 59 percent.

Selection, whether performed at all

Omission ranged from 1 to 38 percent. Hand hygiene after
38, PMI 29, lower extremity edema 21.

Execution was consistent. Selection was not.



Fatigue case: maneuvers not performed

Fatigue case, head-to-toe classes — percent not performed at all

	2023	2024
Assess conjunctivae	70%	85%
Palpates lymph nodes	55%	58%
Light abdominal palpation	62%	74%
Deep abdominal palpation	64%	74%
Lower extremity edema	59%	68%
Heart, 4 positions lying	56%	55%

On a complaint that points to no single organ system, omission rates were high across every maneuver on the checklist.



Refining the core

The criterion we began with

“Performed for every patient, regardless of the presenting problem.”

Difficult to apply: every patient includes an ankle sprain and a rash.

The criterion we are testing

Indicated for any of the twenty most common presenting symptoms.

Bounded and testable. Rated independently by faculty.

7 Core-universal

Indicated irrespective of complaint

5 Core-screening

One maneuver per major system

15 Moves to a cluster

Hypothesis-driven

Provisional. Faculty ratings are being collected.

